

3(a)(i)	any three from: 1 multicellular ; 2 have, specialised cells / tissues / organs ; 3 heterotrophic / described ; 4 <i>ref. to</i> nervous system ; 5 vacuoles / vesicles, are, small / temporary / not permanent ; 6 are eukaryotes / have eukaryotic cells ; 7 detail of eukaryote ; e.g. nucleus / (named) membrane-bound organelle / 80s ribosomes 8 <i>ref. to</i> mobility ;	3
3(a)(ii)	any three from: 1 records / assesses, biodiversity ; 2 so conservation decisions can be made / AW ; 3 example of conservation action 1 ; 4 example of conservation action 2 ; e.g. (IUCN) red list of threatened species / identifying endangered species (CITES) regulation of trade <i>ref.</i> assisted reproduction or <i>ref.</i> captive breeding seed banks place in, zoos / national parks / marine parks / botanical gardens 5 (taxonomic hierarchy is) internationally, recognised / standardised / AW ;	3

Question	Answer	Marks
3(a)(iii)	any two from: 1 (yes because as number of species has increased) biodiversity has increased ; 2 (idea of no because) numbers of new species that are being discovered are greater than the numbers of species becoming extinct ; 3 (idea of no because) does not assess number of individuals in a species ; 4 AVP ; e.g. some species only exist in captivity	2
3(a)(iv)	insects $\frac{1024\ 945}{1438\ 805}(\times 100)$; 71 ; <i>ecf – accept 1 mark if wrong data chosen</i> e.g. $\frac{\text{insects}}{\text{arthropods}} \quad \frac{1024\ 945}{1191770}(\times 100) = 86$ $\frac{\text{arthropods}}{\text{animals}} \quad \frac{1191770}{1438\ 805}(\times 100) = 83$	2
3(b)(i)	any two from: 1 habitat loss / habitat fragmentation / urbanisation / building / deforestation ; 2 insecticides / pesticides / farming / agriculture / fertilizers ; I pollution 3 new / increase in, predators / diseases / parasites ; 4 food shortage / increased competition (for food) ;	2

Question	Answer	Marks
3(b)(ii)	any two from: 1 Bt toxin (only) kills, specific insects / insects that feed on the crops ; 2 reduces use of, insecticides / pesticides ; 3 <i>idea that</i> insects survive to pollinate plants ;	2

Question	Answer	Marks
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